

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 50%

1. Grey Level Transformation

- (a) Explain the purpose of grey level transformation in image enhancement.
- (b) Apply a negative transformation $s = 255 - r$ to the pixel values:

$$r = [50, 100, 150, 200]$$

Compute the transformed values.

2. Log Transformation

- (a) Explain how log transformation enhances low-intensity pixels.

- (b) Given $s = c \log(1 + r)$, where $c = 10$, compute s for:

$$r = [0, 10, 50, 100]$$

3. Histogram Processing

- (a) Explain the role of histogram equalization in contrast enhancement.

- (b) Given a 3-bit image with histogram:

Intensity Frequency

0 2

1 2

2 4

3 8

Compute the cumulative distribution and equalized intensity values.

4. Spatial Filtering (Smoothing)

- (a) Explain the purpose of spatial filtering for smoothing.

- (b) Apply a 3×3 averaging filter to the following region:

1) Grey level Transformation.

Improves the brightness and contrast of an image

Formula : $S = 255 - I$

205, 155, 105, 55

2) Log Transformation.

Enhances dark pixels and compresses bright pixel.

It improves visibility in low intensity regions.

$$F = S \cdot \log(1 + I)$$

0, 24, 39, 46.

3) Histogram equalization

Improves image contrast by spreading pixel

intensity. Makes the image clearer.

$$F = CDF = \text{Cumulative frequency.}$$

$$CDF = 2, 4, 8, 16.$$

$$\text{equalized values} = 1, 2, 4, 7.$$

4) Spatial Filtering.

Smoothing removes noise and reduces unwanted details. It produces a smoother image

$$f = \text{Average} = \text{sum of 9 pixels} \div 9$$

$$\Rightarrow 30.$$

5) Spatial filtering.

Sharpening enhances edges and fine details. It improves image clarity.

$$F = \text{Output} = 4C - (\text{top} + \text{Bottom} + \text{left} + \text{Right}) \\ \Rightarrow 40.$$

6) Frequency Domain Filtering.

Low-pass filter removes high-frequency noise and smooths the image. High pass filter enhance edges and fine details.

$$F = \text{LPF} \rightarrow \text{keep low frequencies only} \\ \Rightarrow [100, 80, 0, 0]$$

7) Thresholding.

Thresholding separates the object from the background using a threshold value. It is used for image seg.

$$F = \text{if } I \geq T \rightarrow 1, \text{ else } 0.$$

$$200 \geq 100 \rightarrow 1$$

8) Edge detection.

Edge detection identifies object boundaries by detecting intensity changes help in feature extraction

$$F = \text{Gradient} = \text{sobelmask} \times \text{image}$$

$$\Rightarrow 80.$$

a) Region - Based Segmentation

Region growing groups neighbouring pixels with similar intensity. It starts from seed pixel.

$$F : [\text{Pixel} - \text{seed}] \leq \text{Threshold}$$

Pixels within ± 5 of 50.

b) Combined enhance noise before segmentation.

Smoothing removes noise before segmentation gives more accurate results.

$$F = \text{Average} \rightarrow \text{threshold.}$$

$$\Rightarrow 48 < 50 \rightarrow 0.$$

11) Power - Law (Gamma)

Gamma correction adjusts image brightness. A gamma value less than 1 brightens image.

$$F = S \cdot c^{\gamma} \quad (c > 1, \gamma > 0.5)$$

$$25 \rightarrow 5 \quad 100 \rightarrow 10, \quad 125 \rightarrow 15.$$

Result: Brightening.

12) Contrast stretching.

Increase the difference between dark and light pixels improves image quality.

$$F = S - (\text{min}) \cdot \frac{(\text{max} - \text{min})}{255}$$

$$50 \rightarrow 0, \quad 100 \rightarrow 128, \quad 150 \rightarrow 255.$$

Better Contrast.

13) Histogram equalization

Histogram equalization improves image contrast using CDF formula: $CDF \rightarrow \text{summing total of freq.}$

$$CDF = 4, 10, 20, 26$$

$$\text{New level} = 2, 6, 12, 15$$

14) Median filtering

Median filtering removes salt-pepper noise while preserving edges. replaces center pixels with median value.

F - median of a pixels.

$$\Rightarrow 50$$

Noise removed.

15) Gaussian smoothing

Reduces noise using weighted average. It gives smoother results than average filter.

$$F : \text{output} = (\text{kernel} \times \text{image}) / k$$

$$\Rightarrow 20$$

Smooth image with reduced noise.